

Question	Answer	Mark	Mark scheme	Additional guidance
22	(12, -8)	P1	for process to rearrange the equation to give y in terms of x , eg $y = \frac{3x-52}{2}$ or $y = \frac{3}{2}x - 26$ or $m = \frac{3}{2}$	Condone an incorrect value for the y intercept ($= c$)
		P1	for process to find gradient of OP , eg $-1 \div \frac{3}{2}$ ($= -\frac{2}{3}$) or $-1 \div [m]$	Where $[m]$ is clearly identified as the gradient of the straight line $3x - 2y = 52$ Allow 0.66(6...) or 0.67 for $\frac{2}{3}$
		P1	(dep on equation of the form $y = \frac{-1}{[m]}x$ for the radius may be implied in subsequent working) for starting to solve $3x - 2y = 52$ with $y = \frac{-1}{[m]}x$ simultaneously to find the value of x or y eg substituting $y = \frac{-2}{3}x$ or $y = \frac{-1}{[m]}x$ into $y = \frac{3}{2}x - 26$ or $3x - 2y = 52$ or an equation of the form $y = \frac{3}{2}x + c$ eg $-\frac{2}{3}x = \frac{3}{2}x - 26$ or $3x - 2\left(-\frac{2}{3}x\right) = 52$ or $-\frac{2}{3}x = \frac{3}{2}x + c$ or $-\frac{1}{[m]}x = \frac{3}{2}x - 26$ or $3x - 2\left(-\frac{1}{[m]}x\right) = 52$ or $-\frac{1}{[m]}x = \frac{3}{2}x + c$	Where $[m]$ is clearly identified as the gradient of the straight line $3x - 2y = 52$ Can be done by elimination Award P1 for a correct method to eliminate x or y : coefficient of x or y the same and correct operator to eliminate selected variable eg $\begin{array}{rcl} 9x - 6y & = & 156 \\ + 4x + 6y & = & 0 \\ \hline 13x & = & 156 \end{array}$ $\begin{array}{rcl} 6x - 4y & = & 104 \\ - 6x + 9y & = & 0 \\ \hline -13y & = & 104 \end{array}$
		A1	cao	